

Scalability on nested request-grant protocols.

We compare VolTRE against the TA-based sampler WORDGEN [wordgen] on a parameterised family that stresses the clock-resource gap between the two formalisms. Consider the following k -level request-grant expression:

$$e_k = \left(\langle r_k \cdot \langle r_{k-1} \cdot \langle \dots \langle r_1 \cdot \langle a^* g \rangle_{\leq 1} \rangle_{\leq 2} \dots \rangle_{\leq k-1} \rangle_{\leq k} \rangle_{\leq k+1} \right)^* \quad (1)$$

Each cycle of e_k models a protocol with k request types $r_k, r_{k-1}, \dots, r_1$ issued in sequence, followed by a grant g within 1 time unit of the final request r_1 . The nested constraints enforce cumulative deadlines: the sub-sequence from r_i must complete within $i+1$ time units. An equivalent timed automaton requires k clocks—one per deadline level—so its state space grows exponentially in the number of clocks, posing a scalability challenge for TA-based sampling approaches. VolTRE, by contrast, processes each $\langle \cdot \rangle_{\leq i}$ layer algebraically through the slice-volume recursion, with only a polynomial overhead per level.

a) Setup.: We fix word length $n = 10$ and vary $k = 1, \dots, 8$. For each k we measure (i) the one-time preprocessing time and (ii) the average per-sample time (cache warm). WORDGEN was given a budget of 3600 s. VolTRE scales to larger n than shown here; we cut off where WORDGEN fails.

b) Results.: Figure 1 shows preprocessing + per-sample time on a logarithmic scale; bars are stacked, with the darker shade showing preprocessing and the lighter shade the per-sample cost (*not* the total over many samples). VolTRE’s per-sample cost grows from ≈ 0.2 s at $k=1$ to ≈ 5.6 s at $k=8$, with preprocessing always under 3 s. WORDGEN is faster for small k (≈ 0.1 s per sample at $k=4$ vs. ≈ 0.3 s for VolTRE), but at $k=6$ its preprocessing alone (≈ 23 s of splitting) already exceeds VolTRE’s combined bar (≈ 4 s), and its per-sample cost rises to ≈ 13 s. WORDGEN runs out of memory at $k=7$. This illustrates a structural advantage of the TRE representation: the exponential blowup in the number of clocks is avoided by VolTRE’s compositional volume computation, at the cost of rejection sampling for intersections that TA-based methods handle natively.

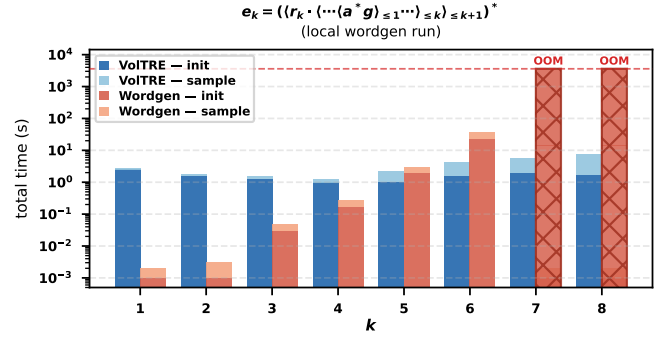


Fig. 1: VolTRE vs. WORDGEN on e_k : preprocessing + per-sample time (log scale, $n = 10$) as k varies. Bars show one sample’s cost, not a total. Hatched bars indicate out-of-memory, dashed line marks the 3600 s budget.